UNITED ARAB EMIRATES MINISTRY OF EDUCATION
الإمارات العربية المتحدة وزارة التربية والتعليم

Al Khalidiya Boys School – Cycle 3

School code 10048 | Dibba Al Hisn | Sharjah Eastern

Grade 12 Physics Final Exam Practice

Term 1 2026–2027 • Set B • With Answers

100 marks | 150 minutes | Calculator allowed

Answer all 20 questions. Choose one option for each MCQ. Show working, units and labelled diagrams for Q17–20. Use g = 9.8 m/s² when needed.

Section A Multiple Choice 60 marks

1. A 1.8 kg brick falls 6.7 m. Its change in gravitational potential energy is: [4 marks]

A. +118 J
B. −12.1 J
C. −118 J
D. 0 J

Reference: p. 20, Practice Problem 33 | **Answer C**

2. If a spring's compression doubles, its stored elastic potential energy becomes: [4 marks]

A. Half as large
B. Four times as large
C. Unchanged
D. Twice as large

Reference: pp. 20–22 | **Answer B**

3. An 8.0 g bullet embeds in a 9.0 kg block at rest. Their final speed is 0.10 m/s. The bullet's initial speed is approximately: [4 marks]

A. 0.113 m/s
B. 113 m/s
C. 1130 m/s
D. 11.3 m/s

Reference: p. 32, Practice Problem 44 | **Answer B**

4. Which feature of thermal radiation could classical physics not explain correctly? [4 marks]

A. The observed energy distribution at short wavelengths
B. That hot objects can emit any radiation
C. That light travels through space
D. The existence of temperature

Reference: pp. 46–48 | **Answer A**

5. A photon has energy 2.03 eV. Its wavelength is approximately: [4 marks]

A. 2.03 nm
B. 611 nm
C. 0.00164 nm
D. 2520 nm

Reference: p. 52, Practice Problem 2 | **Answer B**

6. Light below a metal's threshold frequency strikes it. Increasing intensity alone will: [4 marks]

A. Still fail to eject photoelectrons
B. Reduce the work function to zero
C. Always eject high-energy electrons
D. Increase the frequency of each photon

Reference: pp. 49–51 | **Answer A**

7. At fixed wavelength, the energy of each photon is: [4 marks]

A. Independent of light intensity
B. Zero at low intensity
C. Proportional to intensity
D. Determined by the number of photons only

Reference: pp. 51–56 | **Answer A**

8. What stopping potential is needed for photoelectrons with $KE_{max} = 7.5 \times 10^{-19}$ J? Use $e = 1.60 \times 10^{-19}$ C. [2 marks]

A. 0.213 V
B. 7.5 V
C. 12.0 V
D. 4.69 V

Reference: p. 54, Practice Problem 10 | **Answer D**

9. In Compton scattering, a photon transfers energy to an electron. The scattered photon generally has: [4 marks]

A. Greater energy than before
B. A shorter wavelength
C. Zero momentum at every angle
D. A longer wavelength

Reference: pp. 57–58; p. 58, Check Your Progress 18 | **Answer D**

10. A 7.0 kg bowling ball moves at 8.5 m/s. Its de Broglie wavelength is approximately: [4 marks]

A. 1.11×10^{-25} m
B. 1.11×10^{-35} m
C. 8.5 m
D. 3.94×10^{-32} m

Reference: p. 60, Practice Problem 26 | **Answer B**

11. If the uncertainty in an electron's position is greatly reduced, the minimum uncertainty in its momentum: [4 marks]

A. Becomes exactly its mass
B. Decreases to zero
C. Is always unchanged
D. Increases

Reference: p. 61 | **Answer D**

12. Why did most alpha particles pass nearly straight through the gold foil? [4 marks]

A. The foil contained no atoms
B. Every nucleus attracted them equally
C. Most atomic volume is empty space
D. Alpha particles have no charge

Reference: pp. 62–63 | **Answer C**

13. A low-pressure excited gas generally emits: [4 marks]

A. A line spectrum
B. Only one universal wavelength shared by all gases
C. No radiation
D. Only a continuous spectrum

Reference: pp. 64–66 | **Answer A**

14. A hydrogen atom emits a photon when its electron: [4 marks]

A. Remains in one stationary level
B. Moves from a lower to a higher energy level
C. Gains rest mass at fixed energy
D. Moves from a higher to a lower energy level

Reference: pp. 67–69; p. 74, Check Your Progress 45 | **Answer D**

15. If $r_n = n^2 r_1$ for hydrogen, then r_3/r_1 equals: [4 marks]

A. 3
B. 1/3
C. 9
D. 1/9

Reference: pp. 69–71 | **Answer C**

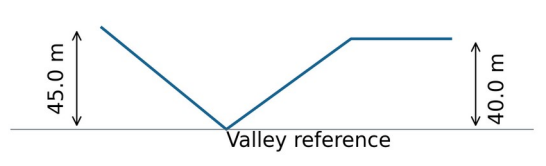
16. Which process is essential to laser operation? [2 marks]

A. Continuous electron spiralling
B. Complete absorption without emission
C. Stimulated emission
D. Random heating alone

Reference: pp. 75–79; p. 79, Check Your Progress 47–50 | **Answer C**

Set B Section B Free Response

17. A skier starts from rest on a hill 45.0 m above a valley and then climbs a second hill 40.0 m above the valley. Ignore friction and pole work. [10 marks]



Reference: p. 28, Practice Problem 41; pp. 23–25

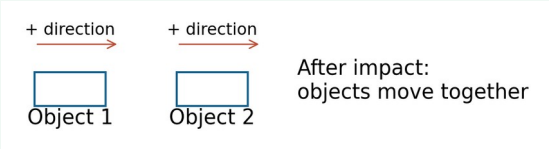
(a) State an appropriate system and the energy conservation equation. [2]
Answer Skier and Earth [1]; $KE_i + PE_i = KE_f + PE_f$ [1].

(b) Calculate speed at the valley floor. [3]
Answer $mg(45) = \tfrac{1}{2}mv^2$ [1]; $v = \sqrt{(2g \times 45)}$ [1] = 29.7 m/s [1].

(c) Find speed at the top of the second hill. [3]
Answer $mg(45) = mg(40) + \tfrac{1}{2}mv^2$ [1]; $v = \sqrt{(2g \times 5)}$ [1] = 9.90 m/s [1].

(d) Do the hill angles affect these speeds? Explain. [2]
Answer No, under the stated assumptions [1]; only vertical height changes determine gravitational energy changes [1].

18. A 91.0 kg hockey player moves at +5.50 m/s. Another 91.0 kg player moving at +8.10 m/s catches the first from behind. They move together after collision. Neglect horizontal external impulse. [10 marks]



Reference: p. 32, Practice Problem 45

(a) Calculate total momentum before collision. [2]

Answer $p_i = 91.0(5.50 + 8.10)$ [1] = 1237.6 kg m/s [1].

(b) Calculate their common final velocity. [2]

Answer $v_f = p_i/(182)$ [1] = +6.80 m/s [1].

(c) Find the initial kinetic energy, final kinetic energy and energy decrease. [4]

Answer $KE_i = ½(91)(5.50^2 + 8.10^2) = 4361.63$ J [1]; $KE_f = ½(182)(6.80^2)$ [1] = 4207.84 J [1]; decrease = 153.79 J [1].

(d) State which quantities are conserved during the collision and justify the classification. [2]

Answer Total momentum and total energy are conserved for the isolated system [1]; KE decreases and they stick, so perfectly inelastic [1].

19. Cesium has work function 1.95 eV and is illuminated by light of wavelength 425 nm. Use hc = 1240 eV nm, h = 4.14 × 10^{-15} eV s and e = 1.60 × 10^{-19} C. [10 marks]

Reference: p. 56, Practice Problem 13; p. 55, Figure 9

(a) Find the incident photon energy. [2]

Answer E = hc/λ [1] = 1240/425 = 2.92 eV [1].

(b) Find maximum photoelectron kinetic energy and stopping potential. [3]

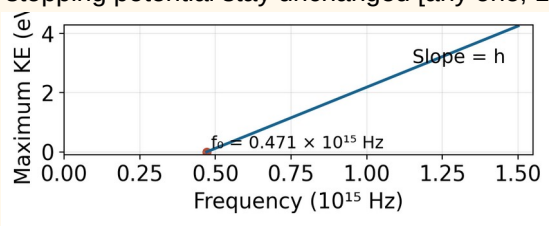
Answer $KE_{max} = E - \varphi$ [1] = 0.968 eV [1]; V_s = 0.968 V [1].

(c) Find the threshold frequency and sketch the KE_{max}–frequency line with the intercept labelled. [3]

Answer $f_0 = \varphi/h = 4.71 \times 10^{14}$ Hz [1]; labelled frequency and KE_{max} axes [1]; rising straight line through (f_0,0), slope h [1].

(d) At fixed 425 nm wavelength, what changes when intensity increases? State one quantity that stays unchanged. [2]

Answer More photoelectrons per second/current increases before saturation [1]; KE_{max}, photon energy and stopping potential stay unchanged [any one, 1].



20. For hydrogen, $E_n = -13.6/n^2$ eV and $r_n = n^2(0.053$ nm). Consider a transition from n = 3 to n = 2. Use h = 4.14 × 10^{-15} eV s and hc = 1240 eV nm. [10 marks]

Reference: p. 72, Practice Problems 33–36; p. 73, Practice Problem 37

(a) Calculate E_3 and E_2. [2]

Answer $E_3 = -13.6/9 = -1.51$ eV [1]; $E_2 = -13.6/4 = -3.40$ eV [1].

(b) Find the photon energy, frequency and wavelength. [4]

Answer $E_{photon} = E_3 - E_2$ [1] = 1.889 eV [1]; f = E/h = 4.56 × 10^{14} Hz [1]; λ = hc/E = 656 nm [1].

(c) Identify the emission series and calculate the radius ratio r_3/r_2. [2]

Answer Balmer series [1]; $r_3/r_2 = 9/4 = 2.25$ [1].

(d) Draw a labelled energy-level diagram and explain why the photon energy is positive although the atom's ΔE is negative. [2]

Answer Levels correctly labelled with downward transition arrow [1]; photon carries away energy lost by atom, $E_{photon} = -\Delta E_{atom} > 0$ [1].

